

Adaptive Air Pressure Regulation System

Design, Development, and Optimization of a Pressure-Controlled Solenoid System Using ESP32

1. Introduction

Maintaining a stable pressure difference between two environments is a common requirement in ventilation systems, enclosures, and controlled airflow applications. This project focuses on the design and implementation of an **adaptive air pressure regulation system** using an ESP32 microcontroller, dual pressure sensors, and a solenoid valve.

The system measures pressure inside and outside an enclosure, computes the differential pressure, and actuates a solenoid valve to correct deviations. Over multiple development iterations, the project evolved from basic pressure sensing into a **fully networked, data-logged, and feedback-driven control system**.

This document explains the **step-by-step progression of the project**, highlighting design decisions, challenges, improvements, and the final optimized solution.

2. System Overview

2.1 Hardware Components

- ESP32 microcontroller
- Two DPS310 digital pressure sensors
 - One measuring outside air pressure
 - One measuring inside air pressure
- Solenoid valve controlled via relay / driver
- Power supply and wiring

2.2 Software Stack

- Arduino framework for ESP32
- DPS310 sensor library
- ESPAsyncWebServer for web interface
- SPIFFS for file storage
- EEPROM for persistent configuration

3. Version 1 – Dual Pressure Sensing and Baseline Calibration

Objective

The first version aimed to **validate pressure sensing hardware** and ensure accurate differential pressure measurement using two DPS310 sensors.

Implementation

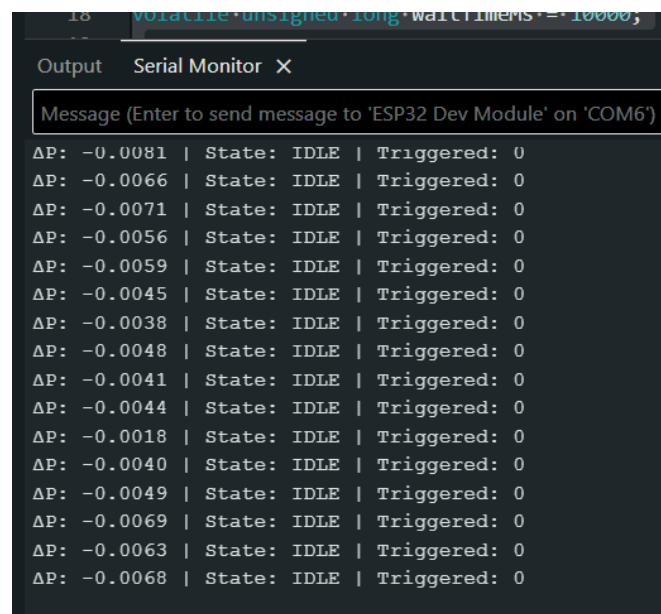
- Two DPS310 sensors were connected over I²C with different addresses.
- High-resolution pressure sampling was enabled for stable readings.
- A baseline calibration was performed at startup to account for sensor offsets.
- Pressure values and their difference were displayed on the serial monitor.

Outcome

This version confirmed:

- Reliable sensor communication
- Stable pressure readings
- Correct calculation of pressure difference

However, there was **no actuator control or automation** at this stage.



```
18 volatile unsigned long waitTime = 1000;
//
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM6')
ΔP: -0.0081 | State: IDLE | Triggered: 0
ΔP: -0.0066 | State: IDLE | Triggered: 0
ΔP: -0.0071 | State: IDLE | Triggered: 0
ΔP: -0.0056 | State: IDLE | Triggered: 0
ΔP: -0.0059 | State: IDLE | Triggered: 0
ΔP: -0.0045 | State: IDLE | Triggered: 0
ΔP: -0.0038 | State: IDLE | Triggered: 0
ΔP: -0.0048 | State: IDLE | Triggered: 0
ΔP: -0.0041 | State: IDLE | Triggered: 0
ΔP: -0.0044 | State: IDLE | Triggered: 0
ΔP: -0.0018 | State: IDLE | Triggered: 0
ΔP: -0.0040 | State: IDLE | Triggered: 0
ΔP: -0.0049 | State: IDLE | Triggered: 0
ΔP: -0.0069 | State: IDLE | Triggered: 0
ΔP: -0.0063 | State: IDLE | Triggered: 0
ΔP: -0.0068 | State: IDLE | Triggered: 0
```

Serial Monitor output showing ΔP

4. Version 2 – Threshold-Based Relay Control

Objective

After validating sensing, Version 2 introduced **basic automatic control** by driving a relay based on pressure difference.

Key Changes

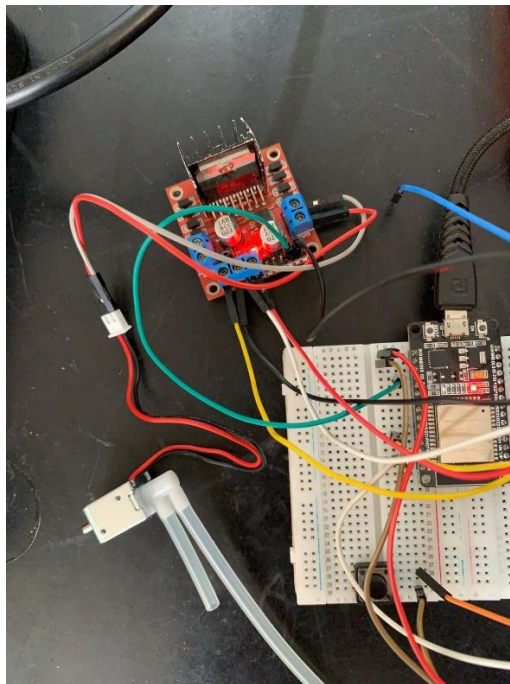
- A relay-controlled solenoid valve was added.
- A pressure threshold determined when the valve should toggle.
- A fixed toggle interval prevented rapid switching.

Observations

While the system reacted to pressure differences, it exhibited:

- Continuous relay toggling
- Pressure oscillations
- Mechanical stress on the solenoid

This version demonstrated that **simple threshold logic was insufficient** for stable control.



Motor Driver module and solenoid valve connection

5. Version 3 – Finite State Machine–Based Control

Objective

Version 3 focused on **stability and deterministic behavior** by replacing simple toggling logic with a **finite state machine (FSM)**.

FSM Design

The solenoid control logic was structured into four states:

- IDLE
- OPEN_PULSE
- CLOSE_PULSE
- WAIT_PHASE

Each correction cycle consisted of:

- Two controlled valve pulses
- Fixed ON and OFF durations
- A long waiting phase to allow pressure stabilization

Improvements

- Eliminated relay chatter
- Reduced overshoot
- Improved system predictability

This version marked the **first reliably stable control behavior**.

6. Version 4 – Networked and Data-Logged Control System

Objective

The goal of Version 4 was to make the system **observable, configurable, and analyzable**.

Major Additions

- ESP32 SoftAP mode for standalone WiFi operation
- Web-based dashboard for real-time monitoring
- Live pressure graph using HTML canvas
- Runtime adjustment of delay between valve cycles

CSV file opened in Excel / Google Sheets

7. Version 5 – Feedback-Driven Final System

Objective

The final version introduced a **human-in-the-loop feedback mechanism**, allowing subjective system performance to be captured and analyzed.

Feedback System

After setting a delay value, the user could provide feedback via the web interface:

- LOW – Delay too short (aggressive response)
- OK – Delay optimal
- HIGH – Delay too long (slow response)

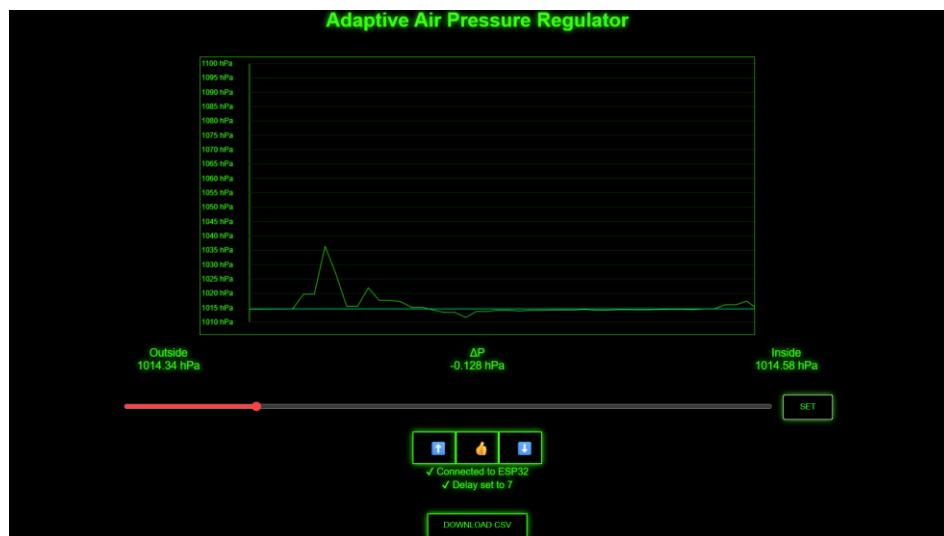
This feedback was:

- Stored in memory
- Logged alongside sensor data in the CSV file
- Initialized as "NULL" at boot to avoid empty entries

Final CSV Structure

time,outside_pressure,inside_pressure,pressure_difference,set_delay,feedback

Each dataset now contained **both objective measurements and subjective evaluation**, enabling optimization.



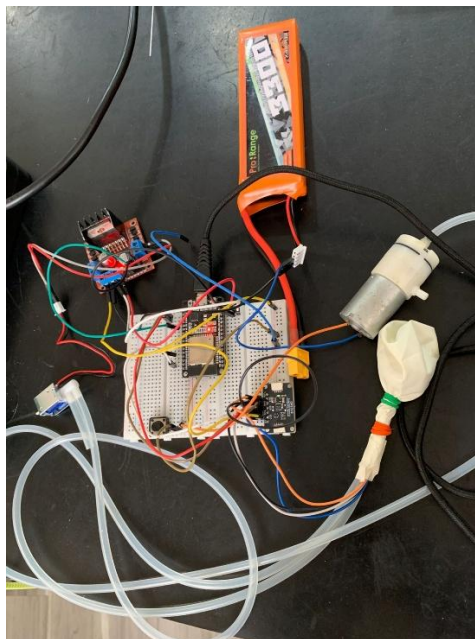
Web UI showing feedback buttons & Feedback confirmation message

8. Data-Driven Optimization

By analyzing CSV logs across multiple runs, it became possible to:

- Compare different delay values
- Correlate feedback with pressure stability
- Identify delay values most frequently rated as “OK”

This transformed the system into an **experiment-driven controller**, rather than a fixed-parameter design.



9. Final System Capabilities

The completed system demonstrates:

- Reliable differential pressure measurement
- Stable solenoid valve control
- Non-blocking real-time operation
- Web-based monitoring and configuration
- Persistent settings across power cycles
- Continuous data logging
- Human feedback integration
- Optimization readiness

10. Project Evolution Summary

Version	Key Achievement
Version 1	Dual pressure sensing & baseline calibration
Version 2	First closed-loop control attempt
Version 3	Stable FSM-based valve control
Version 4	Networked, configurable, logged system
Version 5	Feedback-driven, optimization-ready system

11. Conclusion

This project successfully demonstrates the **design and evolution of an adaptive air pressure regulation system**. Starting from basic sensor validation, the system evolved into a fully featured, data-driven control platform.

By integrating embedded systems, web technologies, persistent storage, and human feedback, the final solution reflects **strong engineering methodology and system-level thinking**.

The approach and architecture developed in this project can be extended to:

- Industrial pressure control
 - Environmental monitoring systems
 - Smart ventilation solutions
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12. Future Scope

Potential future improvements include:

- Automatic delay optimization algorithms
 - Cloud-based data storage
 - Mobile-friendly web interface
 - Advanced control strategies (PID or adaptive control)
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